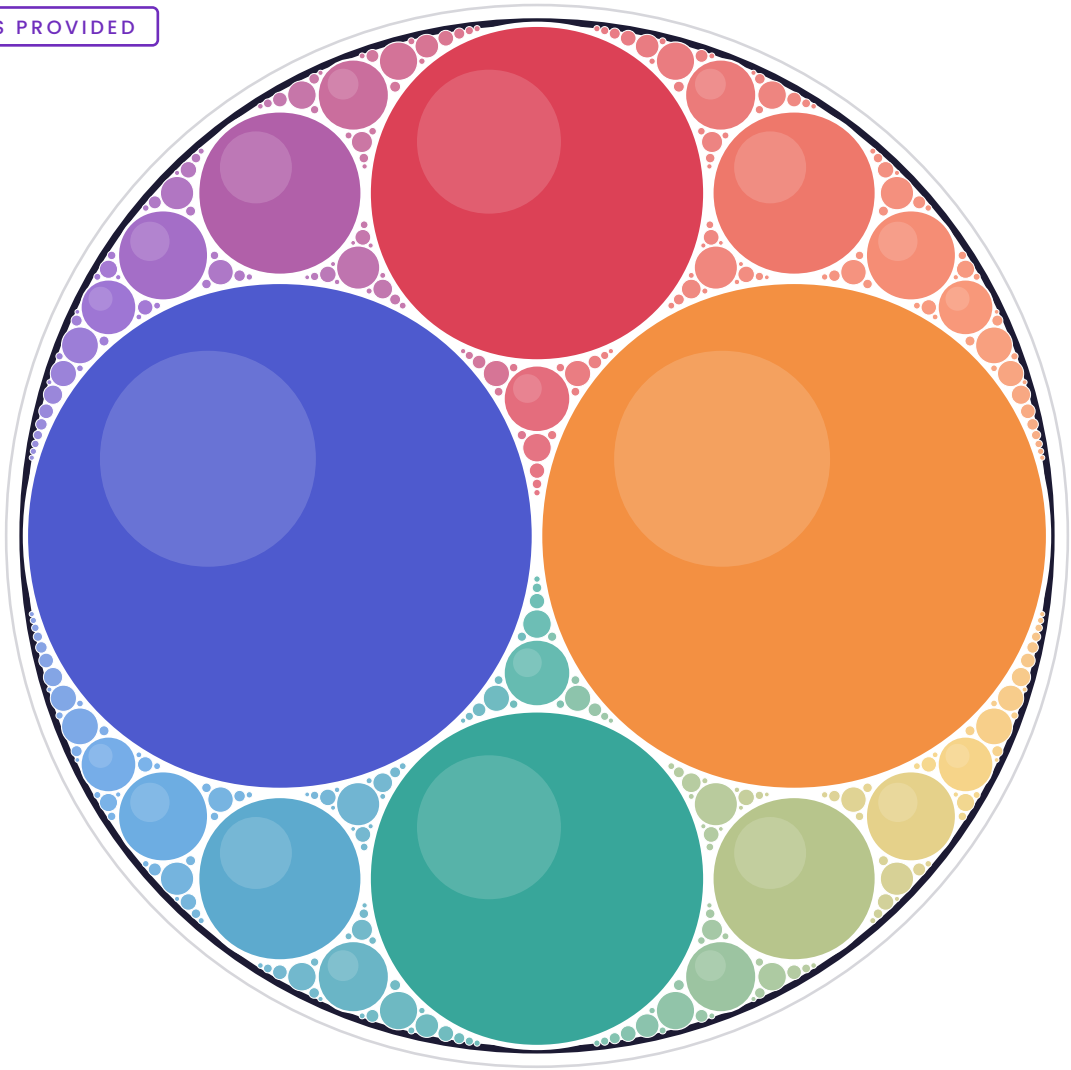
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8 ADVANCED

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
every circle touches three others; every curvature is a whole number

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

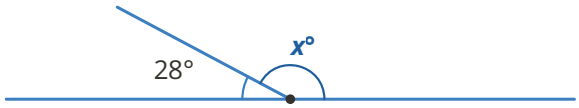
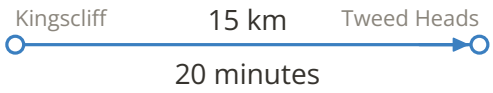
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 4000 mL to litres.	FORMULA $1\ \text{cm}^3 = 1\ \text{mL},\ 1\ \text{L} = 1000\ \text{mL}$ COPY SUBSTITUTE ANSWER
3 Evaluate. 5^3	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 A bus travels from Kingscliff to Tweed Heads. Find the average speed in km/h. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that a six is not rolled. $P(\text{six}) = \frac{1}{6}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

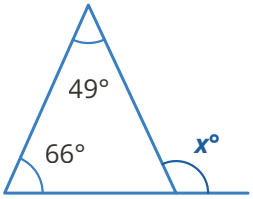
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{4}{5}$ of 35 m	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 A tap fills a tank with 240 L of water in 8 minutes. Find the rate in litres per minute.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 Find the median of these scores. Scores: 7, 3, 9, 5, 8	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

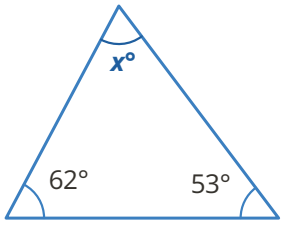
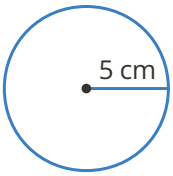
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 0.35	FORMULA Decimal to fraction: write over 10, 100 or 1000, then simplify COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $a = 6$ and $b = 3$. $ab - 4$	FORMULA $ab - 4$ COPY SUBSTITUTE ANSWER
5 A letter is chosen at random from this word. Find $P(\text{M})$. MATHEMATICS	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

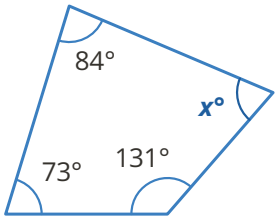
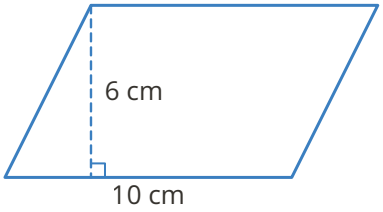
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $15 - 3 \times (2 + 4)$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Solve. $x + 7 = 15$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the mean of these heights. cm: 150, 175, 160, 165, 170	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

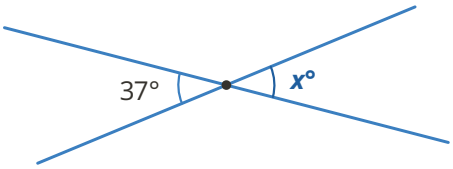
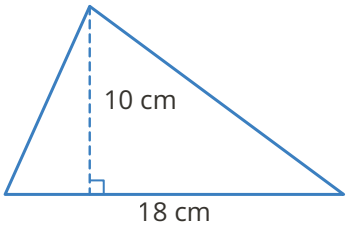
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 10% of $350	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Share 48 marbles in the ratio 3 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 9, 3, 12, 7, 5	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

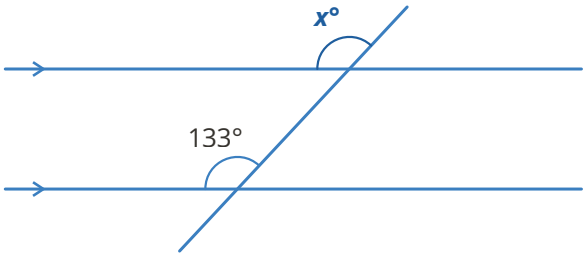
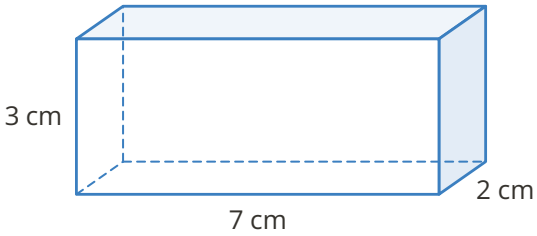
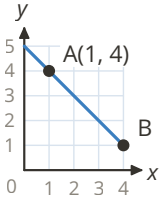
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 A netball shooter scored 21 goals from 25 shots. What percentage of the shots were goals?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $3(x + 5)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 5 - x$. Write the coordinates of B.  	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

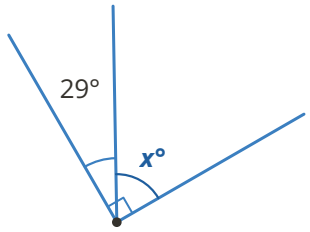
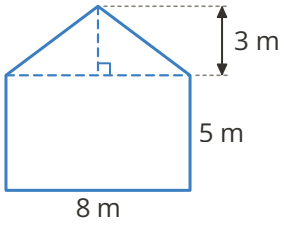
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 3.7×100	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 A movie starts at 19:35 and runs for 1 h 50 min. At what time does it finish? Use 24-hour time.	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 A die was rolled 60 times. Find the experimental probability of rolling a 3. Number: 1, 2, 3, 4, 5, 6 Frequency: 7, 9, 12, 10, 8, 14	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

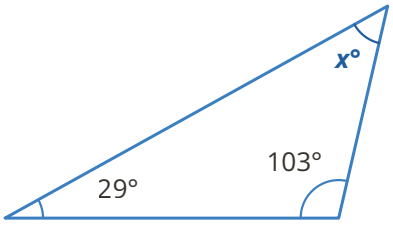
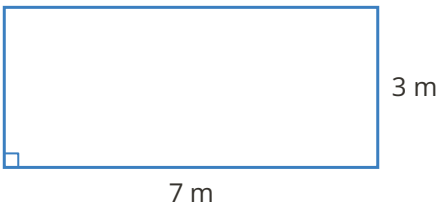
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle.  	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{5}{6}$ of \$42	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $m = 6$ and $n = 7$. $5m - 2n$	FORMULA $5m - 2n$ COPY SUBSTITUTE ANSWER
5 Find the probability that the student does not pass. $P(\text{pass}) = \frac{4}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

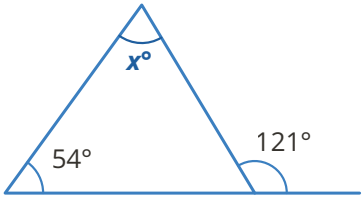
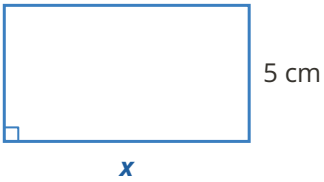
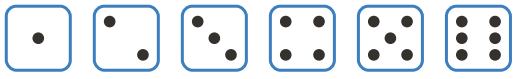
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 The area of this rectangle is 45 cm². Find x. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. $\frac{2}{5}$	FORMULA Fraction to decimal: numerator ÷ denominator COPY SUBSTITUTE ANSWER
4 Simplify. 12 : 18	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a multiple of 3. 	FORMULA $P(\text{E}) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

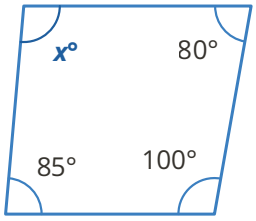
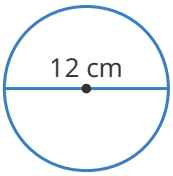
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Calculate. 12% of \$50	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Solve. $6y + 4 = 40$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 8, 3, 5, 6, 8	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

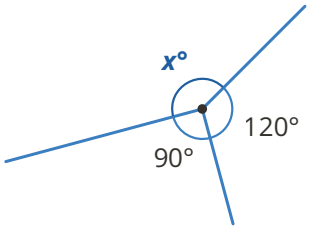
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Convert 7 km to metres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 Evaluate. $-5 \times 4 + 6 \times 3$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 A runner runs from A to B at an average speed of 8 m/s. Find x.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 2 4 4 1 3 2 4 5	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

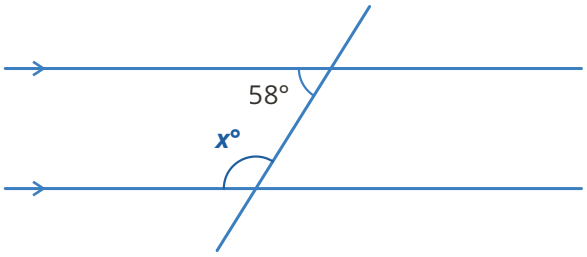
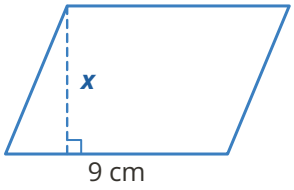
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 The area of this parallelogram is 54 cm². Find x.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt{81} + \sqrt[3]{27}$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Which bag of rice is the better buy? Bag A: 2 kg for $6 Bag B: 5 kg for $20	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 Find the range of these goal tallies. Goals: 5, 11, 8, 2, 9	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

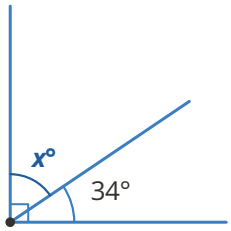
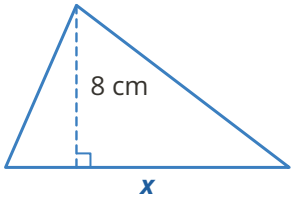
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 The area of this triangle is 56 cm². Find x. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Mia scored 18 out of 24 on a quiz. Write Mia's score as a percentage.	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $6(2 - y)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 A netball goal shooter scored 32 goals from 40 shots. Find the experimental probability of scoring a goal.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

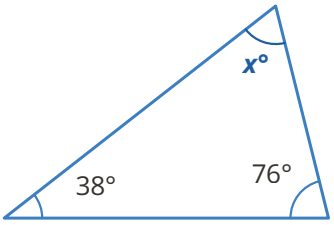
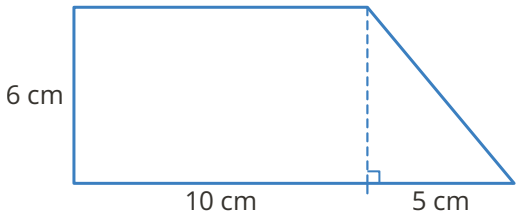
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{1}{6}$ of 54	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $a = 4$ and $b = 5$. $3a + 2b$	FORMULA $3a + 2b$ COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 2x - 2$. x | −1 | 0 | 1 | 2 y | | −2 | 0 |	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

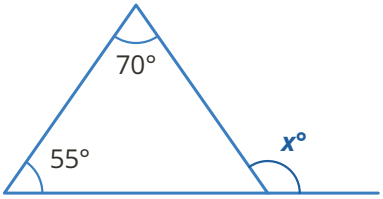
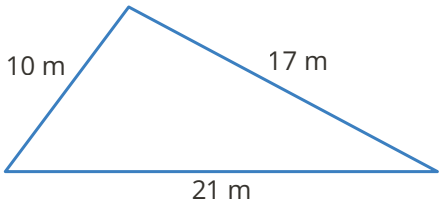
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the perimeter of this triangle.  	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Write as a decimal. 7%	FORMULA Percentage to decimal: percentage ÷ 100 COPY SUBSTITUTE ANSWER
4 Share \$60 in the ratio 1 : 2. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is black. Colour: Black, White Marbles: 7, 3	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

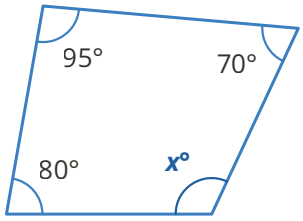
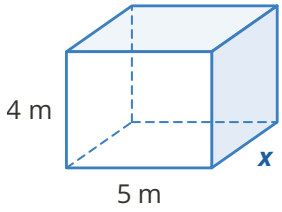
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 The volume of this rectangular prism is 100 m^3. Find x.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Calculate. 30% of 90 L	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Solve. $4x - 7 = 13$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the probability that the marble is not blue. $P(\text{blue}) = \frac{5}{12}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

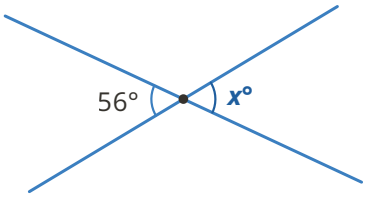
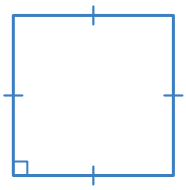
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this square.  8 cm	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Calculate. $82 \div 100$	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Write 12:45 pm in 24-hour time.	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 13, 19, 16, 10, 17	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

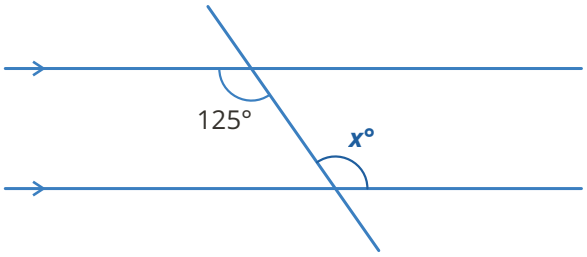
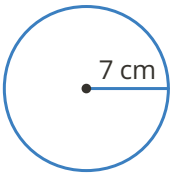
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 19 out of 20	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $7(y + 3)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 5 8 3 8 6 2 8	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

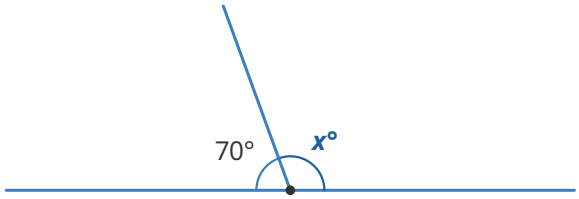
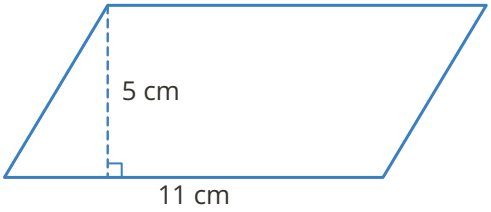
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{7}$ of 56	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Write as a ratio in simplest form. 45 min : 1 h	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 A six came up 15 times in 60 rolls of a die. Find the experimental probability of rolling a six.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

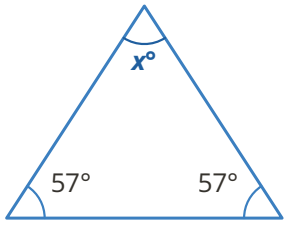
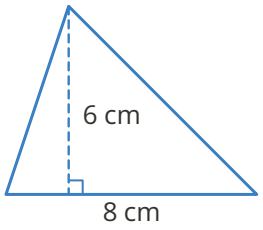
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-20 \div 5 \times 3$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $k = 4$. $20 - 3k$	FORMULA $20 - 3k$ COPY SUBSTITUTE ANSWER
5 Find the range of these test marks. Marks: 58, 72, 65, 49, 70	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

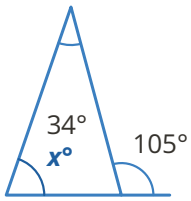
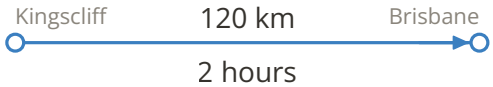
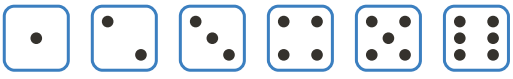
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Convert 2.5 m^2 to square centimetres.	FORMULA $1\ m^2 = 10\ 000\ cm^2$ COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 80%	FORMULA Percentage to fraction: write over 100, then simplify COPY SUBSTITUTE ANSWER
4 A car travels from Kingscliff to Brisbane. Find the average speed.  	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a 5. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

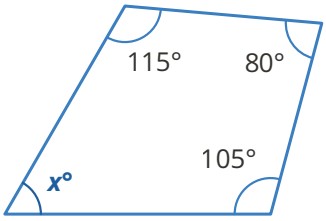
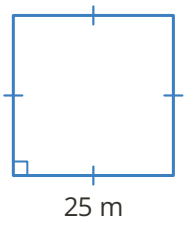
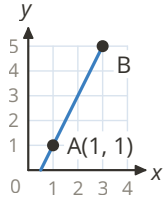
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the perimeter of this square. 	FORMULA $P = 4s$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt{49} + 2^3$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Solve. $m + 12 = 30$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 2x - 1$. Write the coordinates of B. 	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

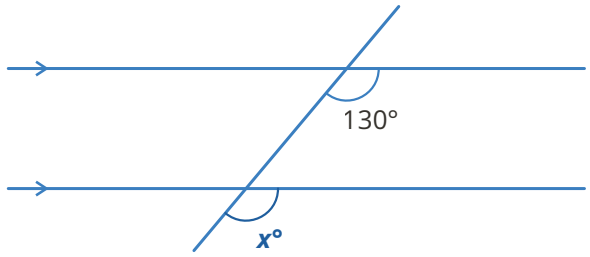
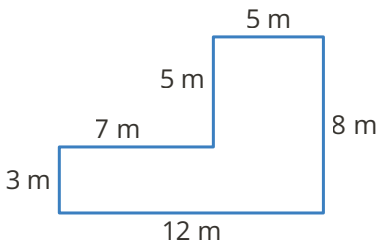
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 20% of 45 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Which bag of potatoes is the better buy? Bag A: 3 kg for $12 Bag B: 5 kg for $15	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 Find the probability that the bus is not late. $P(\text{late}) = \frac{3}{20}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

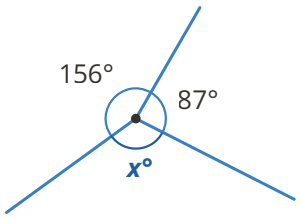
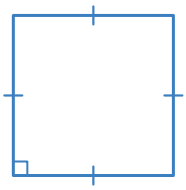
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. 14 out of 40	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $4(m + 9)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mean of these scores. Scores: 30, 25, 35, 20, 40	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

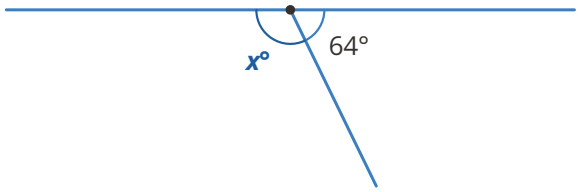
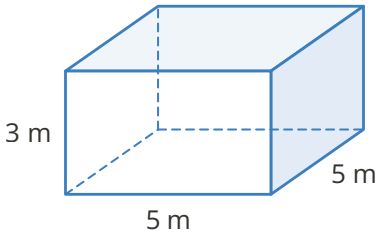
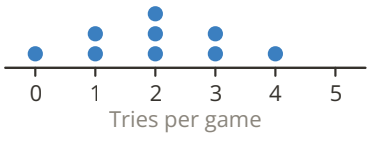
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{7}{10}$ of \$250	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Share 56 shells in the ratio 3 : 4. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 The dot plot shows the tries a rugby league team scored in each game. Find the median. 	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

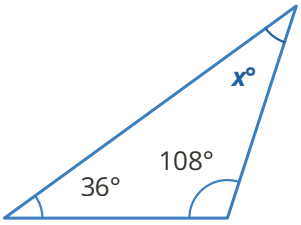
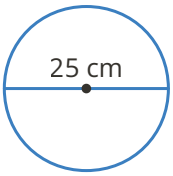
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. $\frac{9}{10}$	FORMULA Fraction to percentage: fraction × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $y = 9$. $5(y - 2)$	FORMULA $5(y - 2)$ COPY SUBSTITUTE ANSWER
5 Heads came up 42 times in 100 tosses of a coin. Find the experimental probability of heads.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

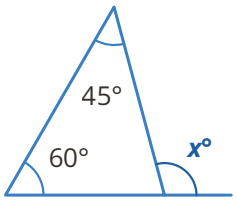
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 0.64×10	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 A bus leaves Pottsville at 08:50. The trip to Tweed Heads takes 45 minutes. At what time does it arrive? Use 24-hour time.	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 Find the range of these marks. Marks: 61, 47, 55, 70, 52	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

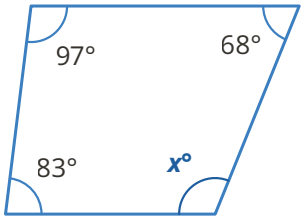
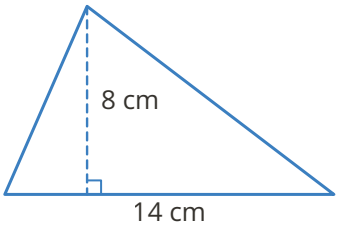
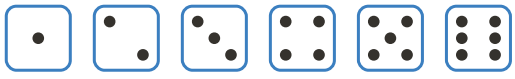
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-3 + 4 \times (-2)$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Solve. $2x + 9 = 25$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a number less than 3. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

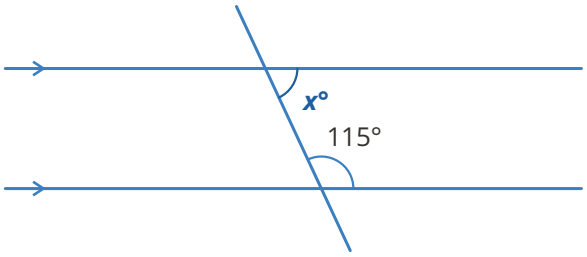
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle.  	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 Calculate. 80% of $45	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Simplify. 14 : 21	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 3x + 2$. x: 0, 1, 2, 3 y: 2, 5, ___, ___	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

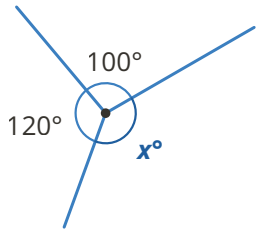
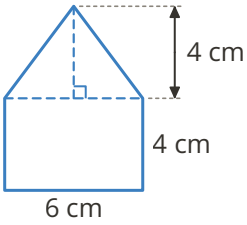
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 1 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 24 of the 30 students in a class walk or ride to school. What percentage is that?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Expand. $2(3x + 5)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mean of these netball scores. Points: 12, 15, 9, 18, 6	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER